

IP Validator Function Project Report

1. Introduction

The IP Validator project is a Python-based networking and cybersecurity project used to verify whether an entered IP address is valid or invalid. The project checks IPv4 address structure according to networking standards.

2. Objectives

- Validate IPv4 addresses
 - Detect invalid IP formats
 - Improve networking knowledge
 - Understand Python functions and string operations
 - Use validation logic in cybersecurity applications
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3. Software Requirements

- Python 3.x
 - Any Python IDE (VS Code, PyCharm, IDLE)
 - Windows/Linux Operating System
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4. Theory

An IP address is a unique numerical identifier assigned to devices on a network. IPv4 addresses contain four octets separated by dots. Each octet must contain values between 0 and 255.

5. Algorithm

1. Accept IP address input from user.
 2. Split the address using dot (.) separator.
 3. Check if total parts are equal to 4.
 4. Verify each part is numeric.
 5. Check if each number is between 0 and 255.
 6. Display Valid or Invalid IP Address.
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6. Python Code

```
def validate_ip(ip):  
    parts = ip.split('.')
```

```
if len(parts) != 4:
    return False

for part in parts:
    if not part.isdigit():
        return False

    num = int(part)

    if num < 0 or num > 255:
        return False

return True
```

7. Sample Output

Enter IP Address: 192.168.1.1
Valid IP Address

8. Advantages

- Easy to understand
 - Useful for networking projects
 - Helps prevent invalid configurations
 - Beginner-friendly cybersecurity project
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9. Limitations

- Supports only IPv4 addresses
 - Does not validate IPv6
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10. Applications

- Network Administration
 - Firewall Systems
 - SOC Monitoring
 - Log Analysis
 - Cybersecurity Tools
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11. Conclusion

The IP Validator project is a simple and effective Python application for validating IPv4 addresses. It improves understanding of networking concepts, Python functions, and cybersecurity fundamentals.

